Para la siguiente gramática, obtener lo que se enlista:

- Factorización por la izquierda
- Conjuntos PRIMERO y SIGUIENTE
- Tabla de análisis sintáctico LL
- Evaluar la cadena (id, (id, (num)), (id))

$$S \rightarrow A \mid B$$

$$A \rightarrow num \mid id$$

$$B \rightarrow (C)$$

$$C \rightarrow S, C \mid S$$

Factorización

$S \to A | B$

$A \to num | id$

$B \to (C)$

$C \to S , C | S$

$C' \to , C | \varepsilon$

$C \to S C'$

$S \to A | B$

$A \to num | id$

$B \to (C)$

$C' \to , C | \varepsilon$

$C \to S C'$

Primeros

$P(S) = \{ num, id, (\}$

$P(A) = \{ num, id \}$

$P(B) = \{ (\}$

$P(C') = \{ , , \varepsilon \}$

$P(C) = \{ num, id, (\}$

Siguientes

$C \to S C'$

$A \quad \alpha \; Z \; \beta$

$P(C') \in \varepsilon$

$S(S) = P(C') \cup S(C)$

$B \to (C)$

$A \quad \alpha \; Z \; \beta$

$P( )) \to \{) \}$

$S(C) = \{) \}$

$S(C') =$

$C \to S C'$

$\alpha \; Z$

$S(C') = S(C)$

$S(A)$

$S \to A$

$S(A) = S(S)$

$C' = \overset{\alpha}{,} \; \overset{Z}{C}$

$S(C) = S(C')$

$S(B) = \{ \$, ,,) \}$

$S \to B$

$A \quad \alpha \; Z$

$S(Z) = S(S)$

$S(S) = \{ \$, ,,) \}$ $\quad S(C) = \{) \}$

$S(C') = \{) \}$ $\quad S(B) = \{ \$, ,,) \}$

$S(A) = \{ \$, ,,) \}$

Tabla

	num	id	(	,	$	)
S	S → A	S → A	S → B			
A	A → num	A → id				
B			B → (C)			
C	C → SC'	C → SC'	C → SC'			
C'				C' → ,C		C' → ε

Pila

	(id,(id,(num)),(id))$
$S	~~(~~id,(id,(num)),(id))$
$B	
$)C~~(~~	
$)C	id,(id,(num)),(id))$
$)C'S	id,(id,(num)),(id))$
$)C'~~A~~	~~id~~,(id,(num)),(id))$
$)C'	~~,~~(id,(num)),(id))$

$\$\ )\ C\ \cancel{,}$	$(id,(num)),(id))\$$
$\$\ )\ C'\ S$	$(id,(num)),(id))\$$
$\$\ )\ C'\ B$	$\cancel{(}id,(num)),(id))\$$
$\$\ )\ C'\ )\ C\ \cancel{(}$	$id,(num)),(id))\$$
$\$\ )\ C'\ )\ C$	$id,(num)),(id))\$$
$\$\ )\ C'\ )\ C'\ S$	$id,(num)),(id))\$$
$\$\ )\ C'\ )\ C'\ \cancel{A}$	$\cancel{id}\cancel{,}(num)),(id))\$$
$\$\ )\ C'\ )\ C\ \cancel{,}$	$(num)),(id))\$$
$\$\ )\ C'\ )\ C'\ S$	$(num)),(id))\$$
$\$\ )\ C'\ )\ C'\ B$	$\cancel{(}num)),(id))\$$
$\$\ )\ C'\ )\ C'\ )\ C\ \cancel{(}$	$num)),(id))\$$
$\$\ )\ C'\ )\ C'\ )\ C'\ S$	$num)),(id))\$$
$\$\ )\ C'\ )\ C'\ )\ C'\ \cancel{A}$	$\cancel{num})),(id))\$$
$\$\ )\ C'\ )\ C'\ )\ \cancel{C'}$	$)),(id))\$$
$\$\ )\ C'\ )\ C'\ \cancel{)}$	$\cancel{)}),(id))\$$

$\$)C')\cancel{C'}$	$),(id))\$$
$\$)C'\cancel{)}$	$\cancel{)},(id))\$$
$\$)C'$	$\cancel{,}(id))\$$
$\$)C\cancel{,}$	$(id))\$$
$\$)C'S$	$(id))\$$
$\$)C'B$	$\cancel{(}id))\$$
$\$)C')C\cancel{(}$	$id))\$$
$\$)C')C$	$id))\$$
$\$)C')C'S$	$id))\$$
$\$)C')C'\cancel{A}$	$\cancel{id}))\$$
$\$)C')\cancel{C'}$	$))\$$
$\$)C'\cancel{)}$	$\cancel{)})\$$
$\$)\cancel{C'}$	$)\$$
$\$\cancel{)}$	$\cancel{)}\$$
$\$$	$\$$ Cadena Válida